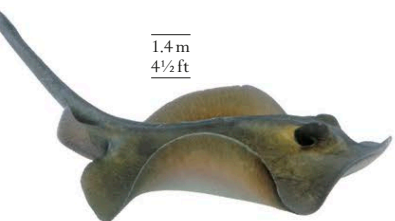


RAYs AND STINGRAYs

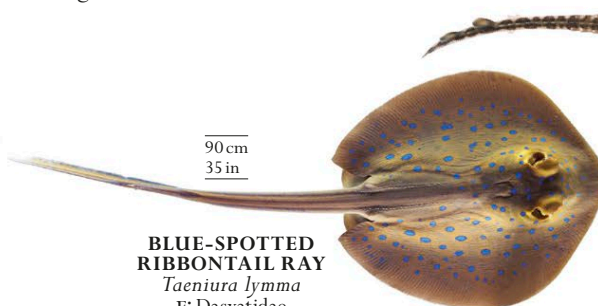
Most rays and skates (Rajiformes) and stingrays (Myliobatiformes) live on the seabed, although they can swim by flapping their wing-like pectoral fins. Their flat, disc-shaped body is well-suited to this way of life. Stingrays have a long, thin tail armed with a venomous spine for protection. Some stingrays spend all or most of the time swimming and a few are found in fresh water.



1.4 m
4½ ft

COMMON STINGRAY
Dasyatis pastinaca
F: Dasyatidae

While most stingrays live in the tropics, the range of this fish is from the Mediterranean into N. Europe, where it lives inshore on sediment.



BLUE-SPOTTED RIBBONTAIL RAY
Taeniura lymma
F: Dasyatidae

Some individuals of this stingray, commonly found on coral reefs in the Indian Ocean and the W. Pacific, have two instead of one venomous tail spine.

tail as long
as body

90 cm
35 in

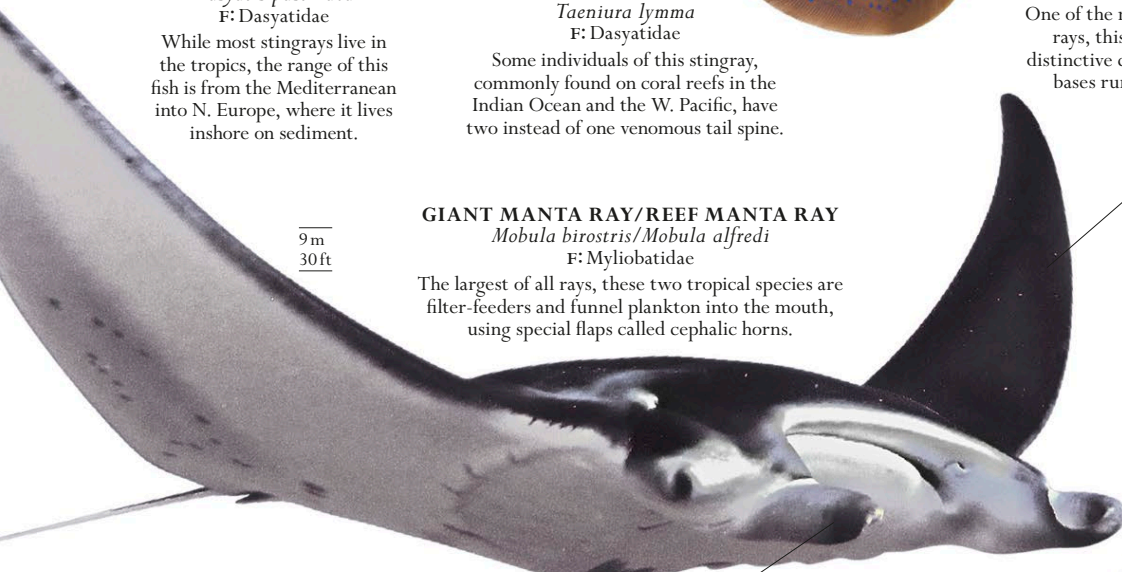
THORNBACK RAY
Raja clavata
F: Rajidae

One of the most common European rays, this species has a row of distinctive curved spines with wide bases running down its back.



BLUE SKATE
Dipturus flossada
F: Rajidae

One of two very similar, large European skates, this species is still sometimes known as *Dipturus batis*. It is distinguished by its long, pointed snout and blue underparts.



9 m
30 ft

GIANT MANTA RAY/REEF MANTA RAY
Mobula birostris/Mobula alfredi
F: Myliobatidae

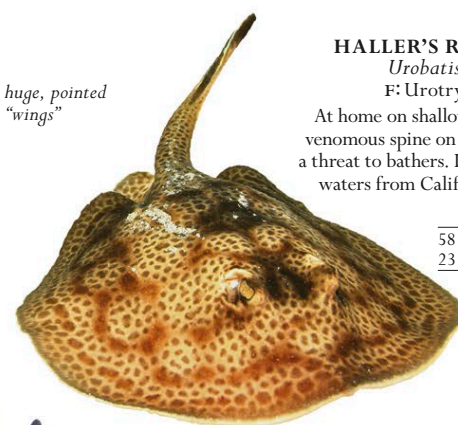
The largest of all rays, these two tropical species are filter-feeders and funnel plankton into the mouth, using special flaps called cephalic horns.

huge, pointed
"wings"

cephalic
horn

SPOTTED EAGLE RAY
Aetobatus narinari
F: Myliobatidae

This tropical stingray noses out molluscs from the sand. It swims well by beating its pectoral fins like a bird's wings.



58 cm
23 in

HALLER'S ROUND RAY
Urobatis halleri
F: Urotrygonidae

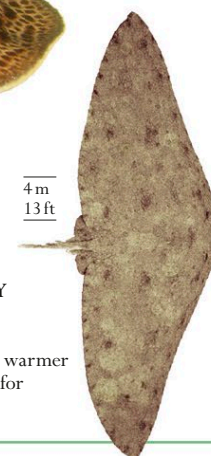
At home on shallow, sandy flats, the venomous spine on this stingray poses a threat to bathers. It is found in coastal waters from California to Panama.



3.3 m
11 ft

SPINY BUTTERFLY RAY
Gymnura altavela
F: Gymnuridae

This ray glides over the seabed in the warmer parts of the Atlantic, searching for invertebrates and fish.



4 m
13 ft

SAWFISH AND GUITARFISH

Sawfish and guitarfish live and hunt on and near the seabed. They have a flattened body, but swim with their tail. Sawfish have a tough, blade-like snout, armed with regular-sized teeth on both edges. Superficially they resemble sawsharks, but, like rays, they have gills on their underside. Recent research places the seven species of sawfish with guitarfish in the order Rhinopristiformes.

ATLANTIC GUITARFISH
Pseudobatos lentiginosus
F: Rhinobatidae

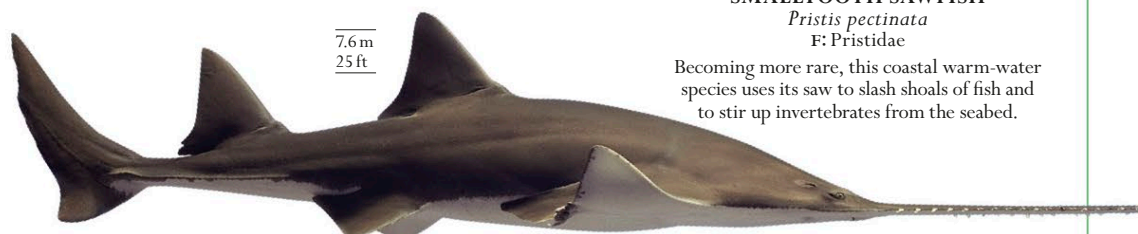
A shovel-shaped snout allows this fish to dig out molluscs and crabs from the sand. Found mainly in the Gulf of Mexico it is now increasingly rare.



75 cm
30 in

SMALLTOOTH SAWFISH
Pristis pectinata
F: Pristidae

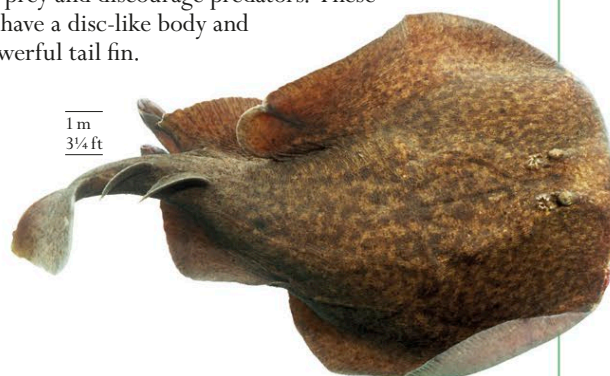
Becoming more rare, this coastal warm-water species uses its saw to slash shoals of fish and to stir up invertebrates from the seabed.



7.6 m
25 ft

ELECTRIC RAYS

The pectoral fins of electric rays, order Torpediniformes, have special organs that produce enough electricity to stun prey and discourage predators. These rays have a disc-like body and a powerful tail fin.



1 m
3¼ ft

MARBLED ELECTRIC RAY
Torpedo marmorata
F: Torpedinidae

When it pounces on a seabed fish, this electric ray can shock its prey with up to 200 volts. Electricity can be produced by even the newborn of this species.